The functor precategory into a category is again a category

The Univalent Foundations Program

hott-book:categories.tex:local-snapshot-20261011; CC BY-SA 3.0 Unported

Editorial note. The counted statement is the author's 'ct:functor-cat' (source lines 353-357, Theorem 9.2.5); the proof environment starts on the next line, so the pairing is by adjacency.

Editorial note. The retained counter values (chapter 9, section 2, theorem offset 4) make the counted theorem print as Theorem 9.2.5, matching the original book numbering taken from the label-number dump of the pinned snapshot.

Editorial note. The single external reference is the transport computation of section 9.1 invoked at source line 372; it is kept as a labelled pointer rather than a context span precisely so that the counted theorem keeps its own section number.

Editorial note. The retained preamble is the chapter's own retained starter preamble for this pinned source.

Editorial scope. One original result of the HoTT Book, counted once: the functor precategory over a category is a category (Theorem 9.2.5).

Allowed standard prerequisites (editorial).

- Functor precategory. B^A has functors as objects and natural transformations as morphisms, with composition and identity defined componentwise.. It is the precategory whose hom-types must be shown to be sets.
- Category. A category is a precategory in which every identity type of the object type is equivalent to the type of isomorphisms via idtoiso.. It is the goal of the theorem.
- Function extensionality and isotoid. Two functions are identified by a homotopy, and isotoid is the inverse of idtoiso on the object type of a category.. They are used to build the candidate inverse of idtoiso at source lines 362-363 and to check the two round trips.

Source reference. Original lemma 9.1.9: sources/hott-categories/book-labelnumbers-first-edition.tex, source lines 1115–1115. Original result cited at source line 372, where the computation of transport along identities in a precategory is used. It is a navigation pointer only: the counted theorem's own author proof is retained line for line and nothing is replaced by it. The number is taken from the label-number dump of the pinned book snapshot. It is kept as a pointer rather

than as a context span because it belongs to another section, and retaining it as context would print it with this section's number.

Theorem 9.2.5. *If A is a precategory and B is a category, then B^A is a category.*

Proof. Let $F, G : A \to B$; we must show that $\mathsf{idtoiso} : (F = G) \to (F \cong G)$ is an equivalence.

To give an inverse to it, suppose $\gamma : F \cong G$ is a natural isomorphism. Then for any $a : A$, we have an isomorphism $\gamma_a : Fa \cong Ga$, hence an identity $\mathsf{isotoid}(\gamma_a) : Fa = Ga$. By function extensionality, we have an identity $\bar{\gamma} : F_0 =_{(A_0 \to B_0)} G_0$.

Now since the last two axioms of a functor are mere propositions, to show that $F = G$ it will suffice to show that for any $a, b : A$, the functions

$$F_{a,b} : \hom_A(a,b) \to \hom_B(Fa, Fb) \qquad \text{and}$$
$$G_{a,b} : \hom_A(a,b) \to \hom_B(Ga, Gb)$$

become equal when transported along $\bar{\gamma}$. By computation for function extensionality, when applied to a, $\bar{\gamma}$ becomes equal to $\mathsf{isotoid}(\gamma_a)$. But by original lemma 9.1.9, transporting $Ff : \hom_B(Fa, Fb)$ along $\mathsf{isotoid}(\gamma_a)$ and $\mathsf{isotoid}(\gamma_b)$ is equal to the composite $\gamma_b \circ Ff \circ (\gamma_a)^{-1}$, which by naturality of γ is equal to Gf.

This completes the definition of a function $(F \cong G) \to (F = G)$. Now consider the composite

$$(F = G) \to (F \cong G) \to (F = G).$$

Since hom-sets are sets, their identity types are mere propositions, so to show that two identities $p, q : F = G$ are equal, it suffices to show that $p =_{F_0 = G_0} q$. But in the definition of $\bar{\gamma}$, if γ were of the form $\mathsf{idtoiso}(p)$, then γ_a would be equal to $\mathsf{idtoiso}(p_a)$ (this can easily be proved by induction on p). Thus, $\mathsf{isotoid}(\gamma_a)$ would be equal to p_a, and so by function extensionality we would have $\bar{\gamma} = p$, which is what we need.

Finally, consider the composite

$$(F \cong G) \to (F = G) \to (F \cong G).$$

Since identity of natural transformations can be tested componentwise, it suffices to show that for each a we have $\mathsf{idtoiso}(\bar{\gamma})_a = \gamma_a$. But as observed above, we have $\mathsf{idtoiso}(\bar{\gamma})_a = \mathsf{idtoiso}((\bar{\gamma})_a)$, while $(\bar{\gamma})_a = \mathsf{isotoid}(\gamma_a)$ by computation for function extensionality. Since $\mathsf{isotoid}$ and $\mathsf{idtoiso}$ are inverses, we have $\mathsf{idtoiso}(\bar{\gamma})_a = \gamma_a$ as desired. □